Kiyan Tavangar Publication List

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First Author or Co-first Author Publications

1. **Tavangar, K.,** et al. (in prep.). *Thalweg: An Updated Flexible Empirical Stream Modeling Framework Applied to ATLAS-Aliqa Uma With Gaia and DES.*

2. **Tavangar, K.,** Johnston, K. V., Hunt, J. A. S., Widmark, A., Kaushik, V. G., & Price-Whelan, A. M. (2026). Phase-Spirals Across Galactic Disks II: Using large-scale "macro-spirals" in phase-spiral amplitude to derive perturbation times. *The Astrophysical Journal, 1010(2),* 126, IOP. https://doi.org/10.3847/1538-4357/aea6b1

3. **Tavangar, K**., Johnston, K. V., Hunt, J. A. S., Widmark, A., Hamilton, C., Petersen, M. S., & Weinberg, M. D. (2026). Phase Spirals across Galactic Disks. I. Exploring Dynamical Influences on Winding. *The Astrophysical Journal*, *1005*(1), 120, IOP. https://doi.org/10.3847/1538-4357/ae7b28

4. Chiti, A., **Tavangar, K.**, Ferguson, P. S., Carballo-Bello, J. A., Senkevich, A. M., Erkal, D., et al. (2025). DELVE-ing into the Milky Way's Globular Clusters: Assessing Extratidal Features in NGC 5897, NGC 7492, and Testing Detectability with Deeper Photometry. *The Astronomical Journal*, *170*(5), 294, IOP. https://doi.org/10.3847/1538-3881/ae07e0

5. **Tavangar, K.**, & Price-Whelan, A. M. (2025). Inferring the Density and Membership of Stellar Streams with Flexible Models: The GD-1 Stream in Gaia Data Release 3. *The Astrophysical Journal*, *988*(1), 45, IOP. https://doi.org/10.3847/1538-4357/addd1c

6. **Tavangar, K.**, Ferguson, P., Shipp, N., Drlica-Wagner, A., Koposov, S., Erkal, D., et al. (2022). From the Fire: A Deeper Look at the Phoenix Stream. *The Astrophysical Journal*, *925*(2), 118, IOP. https://doi.org/10.3847/1538-4357/ac399b

Co-authored Publications with Meaningful Contributions

7. Frankel, N., Semczuk, M., Antoja, T., Chakrabarti, S., Chiba, R., Grand, R. J. J., et al. (2026). Winding, Unwinding, Rewinding the Gaia Phase Spiral. *Publications of the Astronomical Society of the Pacific*, *138*(6), 066001, IOP. https://doi.org/10.1088/1538-3873/ae6bee
   Contribution: I attended the conference which this paper summarizes and wrote the section about the phase spiral mentoring program, which I am leading, along with a few other minor additions.

8. Sanderson, R. E., McKinnon, K. A., Thob, A. C. R., Williams, B., **Tavangar, K**., Pace, A. B., et al. (2026). *A near field guide to Roman's wide-area surveys*. *arXiv e-prints* (arXiv:2603.12981). https://doi.org/10.48550/arXiv.2603.12981
   Contribution: I wrote the section on GD-1 and created Figure 9

9. Widmark, A., **Tavangar, K.**, Kalish, J., Johnston, K. V., & Hunt, J. A. S. (2026). The Phase Spiral's Origin and Evolution: Indications from Its Varying Properties across the Milky Way Disk. *The Astrophysical Journal*, *999*(1), 52, IOP. https://doi.org/10.3847/1538-4357/ae38dd
   Contribution: I had weekly discussions with the lead author about direction of project and I created Figure 11

10. Starkman, N., Nibauer, J., Bovy, J., Webb, J. J., **Tavangar, K.**, Price-Whelan, A., & Bonaca, A. (2025). Stream Members Only: Data-driven Characterization of Stellar Streams with Mixture Density Networks. *The Astrophysical Journal*, *980*(2), 253, IOP. https://doi.org/10.3847/1538-4357/ad94f2
    Contribution: I was leading an analysis with similar objectives but using a different method. The lead authors and I discussed the merits of each method and I provided suggestions for the analysis and the paper.

11. Ferguson, P. S., Shipp, N., Drlica-Wagner, A., Li, T. S., Cerny, W., **Tavangar, K.**, et al. (2022). DELVE-ing into the Jet: A Thin Stellar Stream on a Retrograde Orbit at 30 kpc. *The Astronomical Journal*, *163*(1), 18, IOP. https://doi.org/10.3847/1538-3881/ac3492
    Contribution: I helped create the morphological model used to characterize Jet in this paper.

12. Khullar, G., Gozman, K., Lin, J. J., Martinez, M. N., Matthews Acuña, O. S., Medina, E., et al. (2021). COOL-LAMPS. I. An Extraordinarily Bright Lensed Galaxy at Redshift 5.04. *The Astrophysical Journal*, *906*(2), 107, IOP. https://doi.org/10.3847/1538-4357/abcb86
    Contribution: I searched for strongly lensed objects in the DECaLS data, helped with the source modeling. participated in most discussions for this paper. Note: Authors 3-12 (including **Tavangar, K.)** contributed equally and are listed in alphabetical order

13. Shipp, N., Price-Whelan, A. M., **Tavangar, K.**, Mateu, C., & Drlica-Wagner, A. (2020). Discovery of Extended Tidal Tails around the Globular Cluster Palomar 13. *The Astronomical Journal*, *160*(5), 244, IOP. https://doi.org/10.3847/1538-3881/abbd3a
    Contribution: I quantitatively verified stream significance to support its discovery and created Figure 1 showing the discovered tidal tails

Co-authored Publications with Minor Contributions

14. Chiti, A., Drlica-Wagner, A., Pace, A. B., Cerny, W., Atzberger, K. R., Barbosa, F. O., et al. (2026). *The DECam MAGIC Survey $-$ Mapping the Ancient Galaxy in CaHK: Overview and Summary of Early Science*. *arXiv e-prints* (arXiv:2605.26581). https://doi.org/10.48550/arXiv.2605.26581
    Contribution: I am a MAGIC collaboration member and contributed to discussions when planning the survey

15. Cloonan, A. P., Khullar, G., Napier, K. A., Gladders, M. D., Dahle, H., Rosener, R., et al. (2025). COOL-LAMPS. VIII. Known Wide-separation Lensed Quasars and Their Host Galaxies Reveal a Lack of Evolution in $M_{BH}/M_{\cdot}$ Since $z \sim 3$. *The Astrophysical Journal*, *987*(2), 194, IOP. https://doi.org/10.3847/1538-4357/addabf
    Contribution: I was part of the first class of COOL-LAMPS members and am therefore included on all COOL-LAMPS collaboration papers as a founding member.

16. Mork, S. D., Gladders, M. D., Khullar, G., Sharon, K., Chicoine, N., Cloonan, A. P., et al. (2025). COOL-LAMPS. VII. Quantifying Strong-lens Scaling Relations with 177 Cluster-scale Strong Gravitational Lenses in DECaLS. *The Astrophysical Journal*, *979*(2), 184, IOP. https://doi.org/10.3847/1538-4357/ada24c
    Contribution: I was part of the first class of COOL-LAMPS members and am therefore included on all COOL-LAMPS collaboration papers as a founding member.

17. Klein, M., Sharon, K., Napier, K., Gladders, M. D., Khullar, G., Bayliss, M., et al. (2024). COOL-LAMPS. VI. Lens Model and New Constraints on the Properties of COOL J1241+2219, a Bright $z = 5$ Lyman Break Galaxy and its $z = 1$ Cluster Lens. *The Astrophysical Journal*, *963*(1), 44, IOP. https://doi.org/10.3847/1538-4357/ad22de
    Contribution: I was part of the first class of COOL-LAMPS members and am therefore included on all COOL-LAMPS collaboration papers as a founding member.

18. Aganze, C., Pearson, S., Starkenburg, T., Contardo, G., Johnston, K. V., **Tavangar, K.**, et al. (2024). Prospects for Detecting Gaps in Globular Cluster Stellar Streams in External Galaxies with the Nancy Grace Roman Space Telescope. *The Astrophysical Journal*, *962*(2), 151, IOP. https://doi.org/10.3847/1538-4357/ad159c
    Contribution: I discussed the analysis and results with the lead author and provided suggestions about how to connect this extra-galactic analysis with what has been done in the Milky Way.

19. Zhang, Y., Manwadkar, V., Gladders, M. D., Khullar, G., Dahle, H., Napier, K. A., et al. (2023). COOL-LAMPS. IV. A Sample of Bright Strongly Lensed Galaxies at $3 < z < 4$. *The Astrophysical Journal*, *950*(1), 58, IOP. https://doi.org/10.3847/1538-4357/acc9be
    Contribution: I was part of the first class of COOL-LAMPS members and am therefore included on all COOL-LAMPS collaboration papers as a founding member.

20. Martinez, M. N., Napier, K. A., Cloonan, A. P., Sukay, E., Gozman, K., Merz, K., et al. (2023). COOL-LAMPS. III. Discovery of a 25.″9 Separation Quasar Lensed by a Merging Galaxy Cluster. *The Astrophysical Journal*, *946*(2), 63, IOP. https://doi.org/10.3847/1538-4357/acbe39
    Contribution: I was part of the first class of COOL-LAMPS members and am therefore included on all COOL-LAMPS collaboration papers as a founding member.

21. Sukay, E., Khullar, G., Gladders, M. D., Sharon, K., Mahler, G., Napier, K., et al. (2022). COOL-LAMPS. II. Characterizing the Size and Star Formation History of a Bright Strongly Lensed Early-type Galaxy at Redshift 1.02. *The Astrophysical Journal*, *940*(1), 42, IOP. https://doi.org/10.3847/1538-4357/ac9974
    Contribution: I was part of the first class of COOL-LAMPS members and am therefore included on all COOL-LAMPS collaboration papers as a founding member.

22. Drlica-Wagner, A., Ferguson, P. S., Adamów, M., Aguena, M., Allam, S., Andrade-Oliveira, F., et al. (2022). The DECam Local Volume Exploration Survey Data Release 2. *The Astrophysical Journal Supplement Series*, *261*(2), 38, IOP. https://doi.org/10.3847/1538-4365/ac78eb
    Contribution: I did some observing for the DELVE dataset and am a collaboration member

23. Li, T. S., Ji, A. P., Pace, A. B., Erkal, D., Koposov, S. E., Shipp, N., et al. (2022). S^5: The Orbital and Chemical Properties of One Dozen Stellar Streams. *The Astrophysical Journal*, *928*(1), 30, IOP. https://doi.org/10.3847/1538-4357/ac46d3
    Contribution: I contributed to the S^5 target selection and determining the stream features of some analyzed streams

24. Drlica-Wagner, A., Carlin, J. L., Nidever, D. L., Ferguson, P. S., Kuropatkin, N., Adamów, M., et al. (2021). The DECam Local Volume Exploration Survey: Overview and First Data Release. *The Astrophysical Journal Supplement Series*, *256*(1), 2, IOP. https://doi.org/10.3847/1538-4365/ac079d
    Contribution: I did some observing for the DELVE dataset and am a collaboration member

25. Stringer, K. M., Drlica-Wagner, A., Macri, L., Martínez-Vázquez, C. E., Vivas, A. K., Ferguson, P., et al. (2021). Identifying RR Lyrae Variable Stars in Six Years of the Dark Energy Survey. *The Astrophysical Journal*, *911*(2), 109, IOP. https://doi.org/10.3847/1538-4357/abe873
    Contribution: I searched for main sequence stellar populations associated with RRL clumps

26. Mau, S., Cerny, W., Pace, A. B., Choi, Y., Drlica-Wagner, A., Santana-Silva, L., et al. (2020). Two Ultra-faint Milky Way Stellar Systems Discovered in Early Data from the DECam Local Volume Exploration Survey. *The Astrophysical Journal*, *890*(2), 136, IOP. https://doi.org/10.3847/1538-4357/ab6c67
    Contribution: I did some observing for the dataset and worked on the satellite search tools used here